CHAPTER 5 INTEGRALS

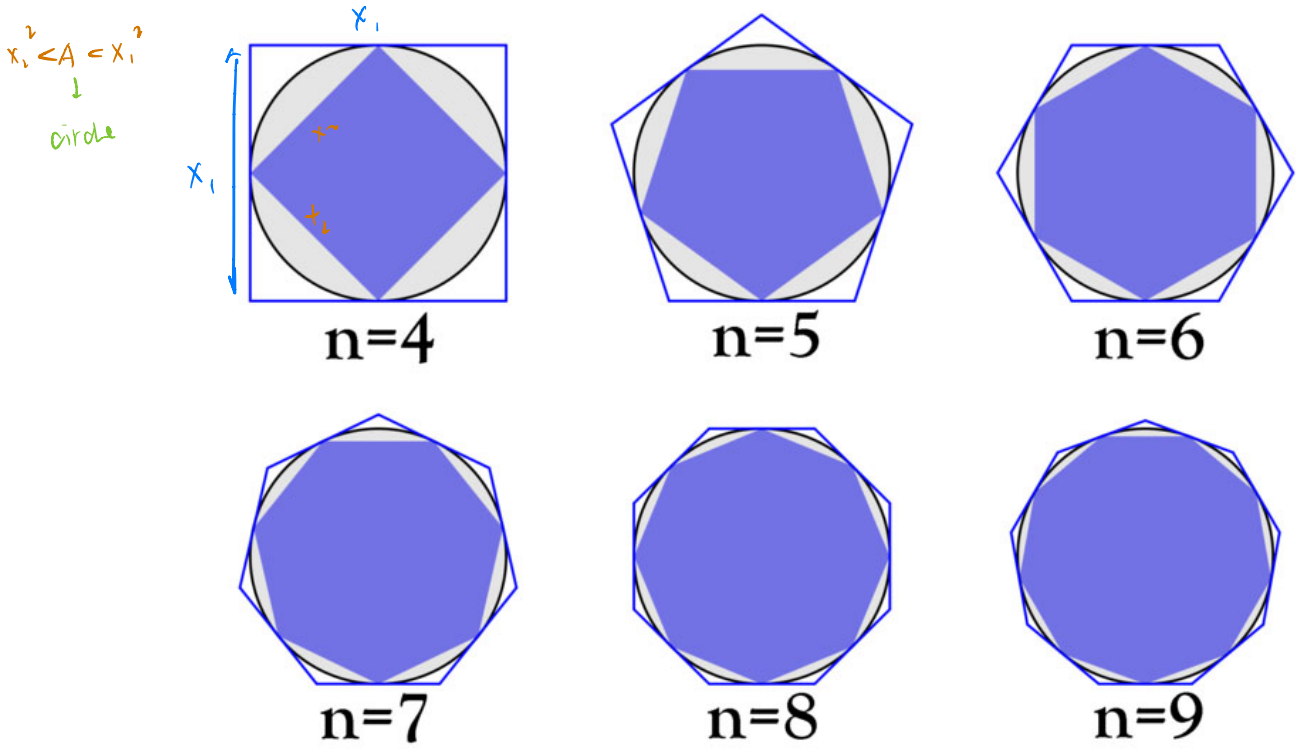
5.1 Areas and Distances

- How do we find the area of a circle?

  Over 2000 years ago, Archimedes, a famous Greek mathematician, provided the key to a solution.

  Consider a sequence of **inscribed polygons** that approximate the circle with greater and greater accuracy.

  Archimedes also considered a sequence of **circumscribed polygons** that approximate the circle with greater and greater accuracy.



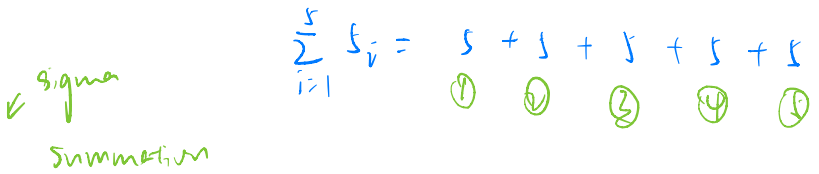
Let $A(P_n)$ be the area of the polygon with n sides.

$$A(\text{circle}) = \lim_{n\to\infty} A(P_n)$$

- **Σ Notation**

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + \ldots + a_{n-1} + a_n$$

- **Properties of Σ**



- $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$
- $\sum_{i=1}^{n} (a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$

- **Some Special Sum Formulas**

1. $\sum_{i=1}^{n} 1 = 1 + 1 + 1 + \cdots + 1 = n$
2. $\sum_{i=1}^{n} i = 1 + 2 + 3 + \cdots + n = \dfrac{n(n+1)}{2}$
3. $\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$
4. $\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2$
5. $\sum_{i=1}^{n} i^4 = 1^4 + 2^4 + 3^4 + \cdots + n^4 = \dfrac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$

Example 5.1. Find a formula for

$$R_n = \frac{1}{n}\left(\frac{1}{n}\right)^2 + \frac{1}{n}\left(\frac{2}{n}\right)^2 + \frac{1}{n}\left(\frac{3}{n}\right)^2 + \cdots + \frac{1}{n}\left(\frac{n}{n}\right)^2$$

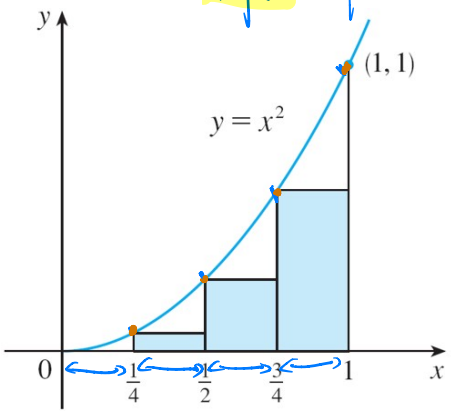
Riemann Sum

- **Areas**

Example 5.2. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using four rectangles and left endpoints.

Area of rectangles < Area under the curve

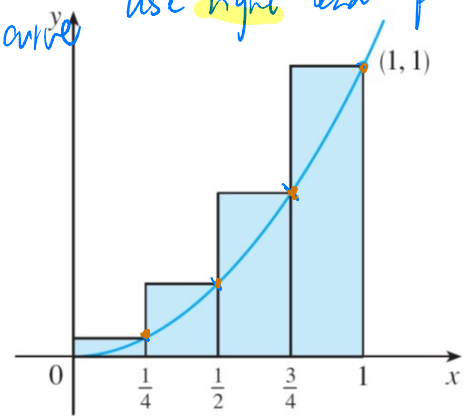
use left end points



Example 5.3. Estimate the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using four rectangles and right endpoints.

Area of rectangles > area under the curve

use right end points



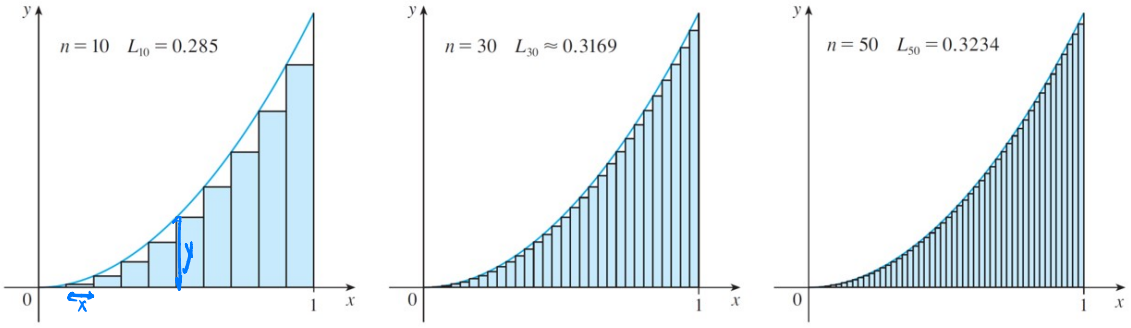
Example 5.4. Find the area of the region bounded by the curve $y = x^2$, the line $x = 1$ and the x-axis using n rectangles and right endpoints with $n \to \infty$.

as $n \uparrow$ $L_n \Rightarrow R_n$

n	L_n	R_n
10	0.2850000	0.3850000
20	0.3087500	0.3587500
30	0.3168519	0.3501852
50	0.3234000	0.3434000
100	0.3283500	0.3383500
1000	0.3328335	0.3338335

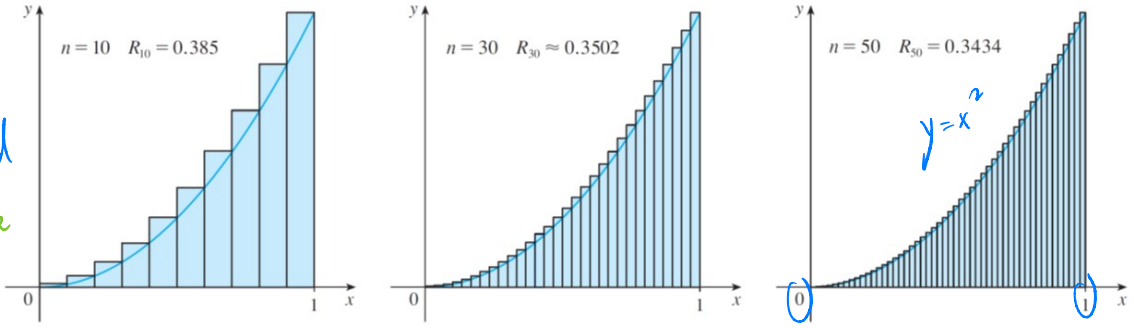
use left end

under estimate



use right end

over estimate



Exercises 5.1 5(a)(b) and find the area by using n rectangles and right endpoints with $n \to \infty$ (area = 6), 17.

5.2 The Definite Integral

- **Riemann Sums**

  Consider a function defined on a closed interval $[a, b]$. Divide $[a, b]$ into n equal subintervals by means of points $a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b$, where

$$x_i = a + i\Delta x$$

  and

$$\Delta x = \frac{b-a}{n}.$$

  On each subinterval $[x_{i-1}, x_i]$, pick an arbitrary point x_i^*, which we call a sample point of the interval.

Let f be a function that is defined on the closed interval $[a, b]$. If

$$\lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

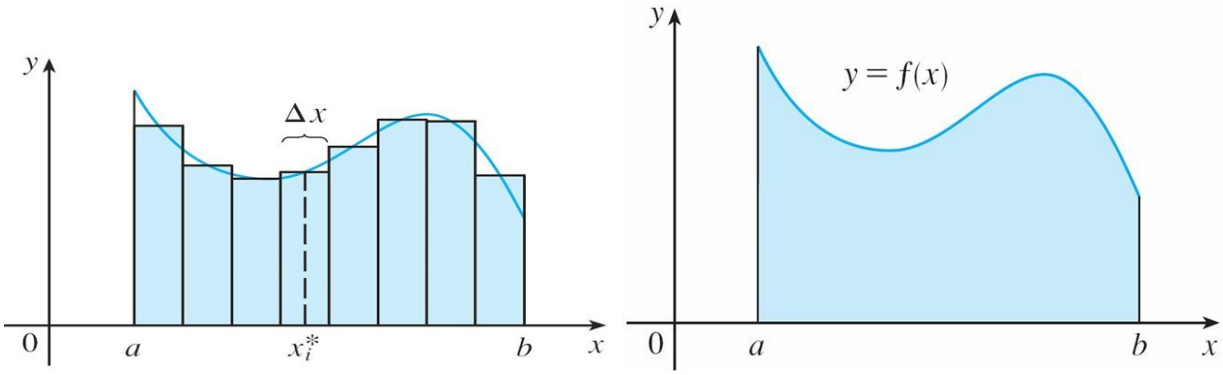
exists, we say that f is **integrable** on $[a, b]$, and denote this limit by $\int_a^b f(x)dx$, called the **definite integral** of f from a to b. Namely,

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

$\leftarrow$ width of rectangle

$\uparrow$ y : height of rectangle

a and b are called the **lower limit** and **upper limit** of the integral, $f(x)$ is called the **integrand**.



Example 5.5. Evaluate $\int_{-1}^{3} \left(2x^2 - 8x\right)\, dx$ by using the definition of the integral.

$\Delta x =$

$x_i =$

$f(x_i^*)\Delta x =$

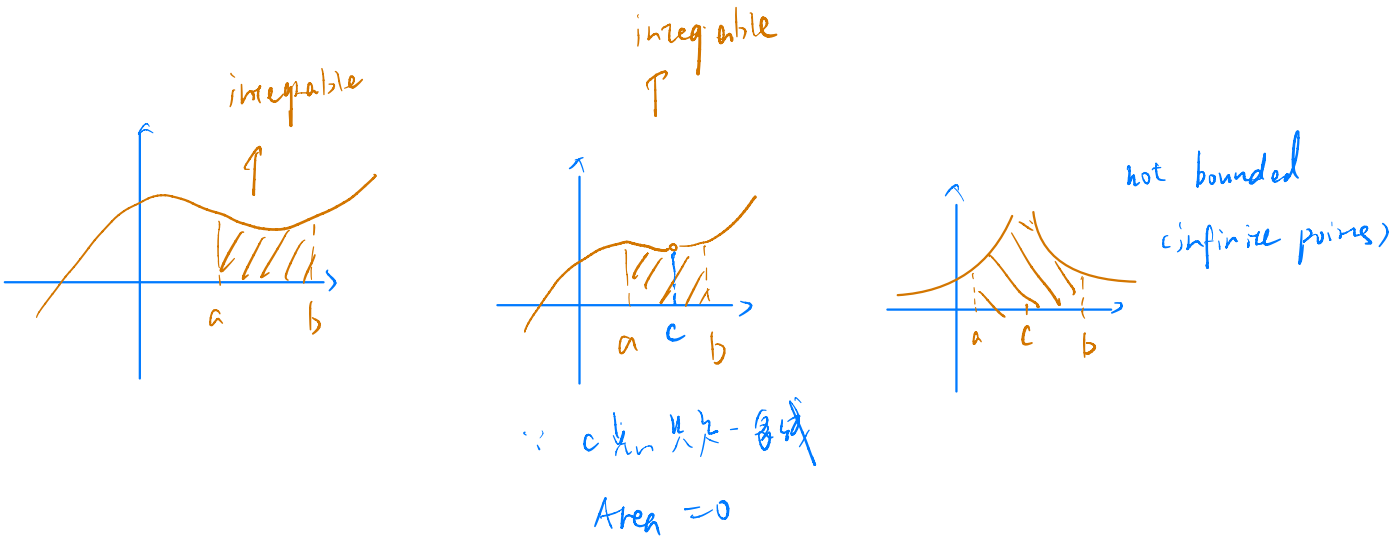
$\sum_{i=1}^{n} f(x_i^*)\Delta x =$

$\int_{-1}^{3} (2x^2 - 8x)dx =$

- **Theorem**

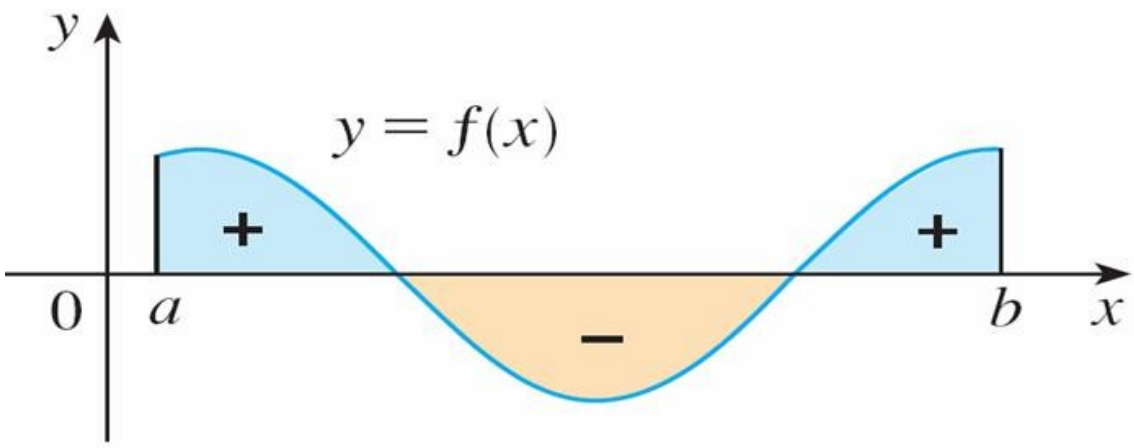
> If f is bounded on $[a, b]$ and if it is continuous there except at finite number of points, then f is integrable on $[a, b]$.

> If f is continuous on $[a, b]$, then it is integrable on $[a, b]$.



- **Geometric Interpretation**

$$\int_a^b f(x)dx = A_{\text{up}} - A_{\text{down}}$$



Example 5.6. An object at the origin at time $t = 0$ has velocity, measured in meters per second,

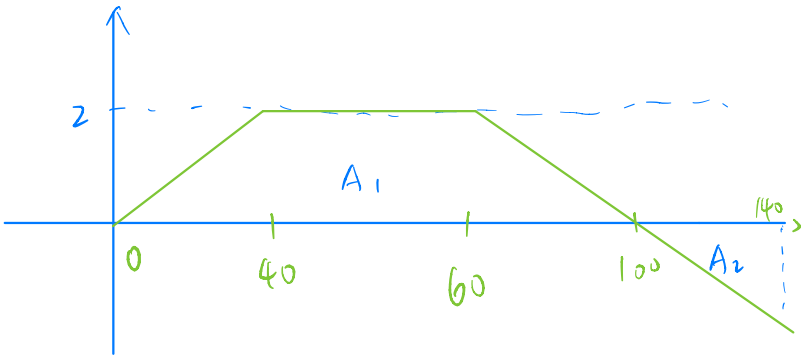
$$v(t) = \begin{cases} \dfrac{t}{20}, & 0 \le t \le 40 \\ 2, & 40 < t \le 60 \\ 5 - \dfrac{t}{20}, & t > 60 \end{cases}$$

Sketch the velocity curve. Express the position of the object at $t = 140$ s as a definite integral and evaluate it using formulas from plane geometry.

$0 \sim 100$: curve above x-axis

$100 \sim 140$ curve below x-axis

$A = \int_0^{100} v(t)\,dt - \int_{100}^{140} v(t)\,dt$



Example 5.7. Evaluate $\int_0^3 \sqrt{9 - x^2}\,dx$.

semi circle

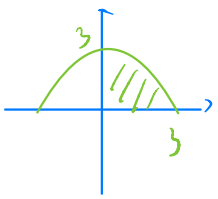
$x^2 + y^2 = 3^2$

$y = \sqrt{9 - x^2}$

$= \frac{1}{4} \times$ Area of circle with radius 3

$= \frac{1}{4}\pi(3)^2$

$= \frac{9}{4}\pi$

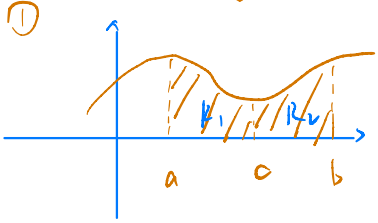


- **Properties of the Definite Integral**

- $\int_a^a f(x)dx = 0.$
- $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx.$ 与 c 的相对位置无关 (只要 integrable)
- $\int_a^b f(x)dx = -\int_b^a f(x)dx.$

↓ $a<c<b$ ① ; $a<b<c$ ②

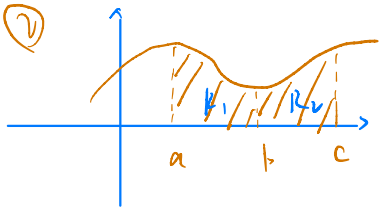
① $a<c<b$



$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

$K = K_1 + K_2$

② $a<b<c$



$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$

$= \int_a^c f(x)dx - \int_b^c f(x)dx$

$= -\int_b^c f(x)dx$

Example 5.8. Given that $\int_0^{10} f(x)dx = 17$, $\int_0^8 f(x)dx = 12$, find $\int_8^{10} f(x)dx$.

对 a、b、c

$\int_8^{10} f(x)dx = \int_8^0 f(x)dx + \int_0^{10} f(x)dx$

$= -\int_0^8 f(x)dx + \int_0^{10} f(x)dx$

$= -12 + 17$

$= 5$

- **Linearity of the Definite Integral**

If f and g are integrable on $[a,b]$ and k is a constant, then kf and $f \pm g$ are integrable and

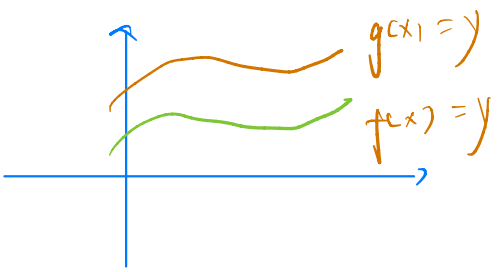
- $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
- $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

- **Comparison Property**

Area under the curve larger as function

If f and g are integrable on $[a,b]$ and if $f(x) \leq g(x)$ for all $x \in [a,b]$, then

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx$$



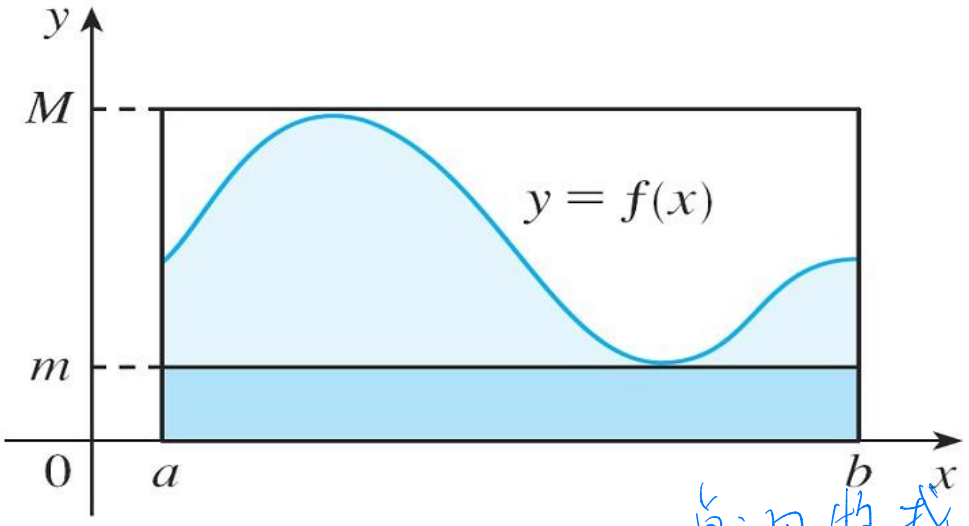
- **Boundedness Property**

length

constant

If f is integrable on $[a, b]$ and $m \le f(x) \le M$ for all x in $[a, b]$, then

$$m(b-a) \le \int_a^b f(x)dx \le M(b-a)$$



Example 5.9. Give an estimate of $\int_0^1 e^{-x^2}dx$.

找到值域

maximum M:

minimum: m

$\frac{1}{e} \le e^{-x^2} \le 1$

Let $f(x) = e^{-x^2}$

$f'(x) = -2xe^{-x^2}$ always > 0

$f(x)$ always decrease when $0 < x < 1$

$\therefore$ when $x = 0$, there is a maximum

$x = 1$ minimum

According to the Boundness Property,

$\frac{1}{e}(1-0) \le \int_0^1 e^{-x^2}dx \le 1(1-0)$

$0.3679 \le \int_0^1 e^{-x^2}dx \le 1$

Exercises 5.2 17, 19, 23, 35, 37, 39, 41, 47, 49, 55, 57.

5.3 The Fundamental Theorem of Calculus

- Calculus: Differential Calculus and Integral Calculus.
- Differential Calculus: arose from the tangent problem.
- Integral Calculus: arose from the area problem.
- Newton's mentor, Isaac Barrow $(1630 - 1677)$, discovered that these two problems are actually closedly related.

- **First Fundamental Theorem of Calculus** (FTC)

Let f be continuous on the closed interval $[a, b]$ and let x be a variable point in (a, b). Then

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

only x (eg x^2 x+c)

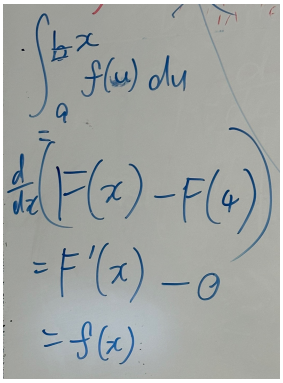
Differentiate an integration of f(t) will get the original f(t) => differentiation and integral are inverse

constant

Example 5.10. Find $\dfrac{d}{dx}\left[\displaystyle\int_1^x t^3 dt\right]$.

$= t^3$

Example 5.11. Find $\dfrac{d}{dx}\left[\displaystyle\int_x^4 \tan^2 u \cos u du\right]$, $\dfrac{\pi}{2} < x < \dfrac{3\pi}{2}$.



$= \frac{d}{dx}\left[-\int_4^x \tan^2 u \cos u du\right]$

$= \frac{d}{dx}\left[-\int_4^x \tan^2 u \cos u du\right]$ replace u with x (why?)

$= -\tan^2 x \cos x$

Example 5.12. Find $D_x\left[\displaystyle\int_x^{x^2} \sqrt{3t^2 - 1} dt\right]$.

any number inside the domain

$= D_x\left[\int_x^1 \sqrt{3t^2-1}\,dt + \int_1^{x^2} \sqrt{3t^2-1}\,dt\right)$

$= D_x\left[-\int_1^x \sqrt{3t^2-1}\,dt + \int_1^{x^2} \sqrt{3t^2-1}\,dt\right)$

$= -\sqrt{3t^2-1} + \frac{d}{dx}\left[\int_1^{x^2} \sqrt{3t^2-1}\,dt\right)$

$= -\sqrt{3t^2-1} + \frac{du}{dx}\frac{d}{du}\int_1^u \sqrt{3u^2-1}\,dt$

$+ 2x\sqrt{3x^4-1}$

$\frac{du}{dx} = 2x$

let $u = x^2$

$\frac{d}{dx} = \frac{du}{dx}\cdot\frac{d}{du}$

?

$\frac{d}{dx}(x) = 1 \Rightarrow$ antiderivative of 1 is x

- **Second Fundamental Theorem of Calculus**

Let f be continuous on $[a, b]$, and let F be any antiderivative of f on $[a, b]$. Then

$$\int_a^b f(x)dx = F(b) - F(a)$$

$\frac{d}{dx}(x+3) = 1 \Rightarrow$ antiderivative of 1 is $x+3$

Example 5.13. Show that $\int_a^b c\,dx = c(b-a)$.

$\int_a^b c\,dx = [cx]_a^b = cb - ca$

$F(x)$

Example 5.14. Evaluate $\int_a^b x\,dx$.

$= \left[\frac{x^2}{2}\right]_a^b = \frac{b^2}{2} - \frac{a^2}{2}$

Example 5.15. Given that r is a rational number, $r \neq -1$, evaluate $\int_a^b x^r\,dx$.

$\int_a^b x^r\,dx = \left[\frac{x^{r+1}}{r+1}\right]_a^b \times \left[\frac{x^{r+1}}{r+1}\right]_a^b = \frac{b^{r+1}}{r+1} - \frac{a^{r+1}}{r+1}$

Example 5.16. Evaluate $\int_{-1}^{2} (4x - 6x^2)\,dx$.

$= \left[\frac{4x^2}{2} - \frac{6x^3}{3}\right]_{-1}^{2}$

$= \left[2x^2 - 2x^3\right]_{-1}^{2}$

Example 5.17. Evaluate $\int_1^8 \left(x^{\frac{1}{3}} + x^{\frac{4}{3}}\right) dx.$

Example 5.18. Find $\int_0^x 3\sin t dt.$

$= \left[-3\cos t\right]_0^x$

$= -3\cos x + 3$

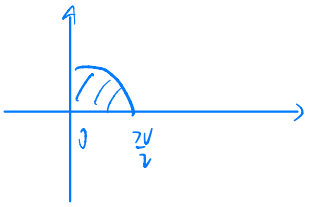
Example 5.19. Evaluate $\int_1^3 e^x dx.$

$= \left[e^x\right]_1^3$

Example 5.20. Evaluate $\int_3^6 \frac{1}{x} dx$.

Example 5.21. Find the area under the curve $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$.

between curve and the x-axis

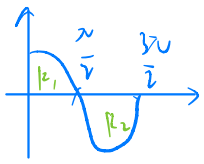


$$\text{Area} = \int_0^{\frac{\pi}{2}} \cos x \, dx$$

$$= [\sin x]_0^{\frac{\pi}{2}}$$

$$= 1$$

between the curve from $x = \frac{\pi}{2}$ to $x = \pi$



$$\text{Area} = \int_0^{\frac{\pi}{2}} \cos x \, dx - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos x \, dx$$

(Δ: area under the curve)

$$= [\sin x]_0^{\frac{\pi}{2}} - [\sin x]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$$

$$= 1 - [(-1) - 1]$$

$$= 3$$

Exercises 5.3 7, 11, 13, 17, 21, 25, 29, 31, 37, 41, 43, 47, 61, 67, 75.

5.4 Indefinite Integrals and the Net Change Theorem

- The notation $\int f(x)dx$ is traditionally used for an antiderivative of f and is called an **indefinite integral**.

$\int f(x)dx = F(x)$ means $F'(x) = f(x)$

- **Standard Integral Formulas**

1. $\int kdu = ku + C$
2. $\int u^r du = \dfrac{u^{r+1}}{r+1} + C \quad (r \neq -1)$
3. $\int \dfrac{1}{u} du = \ln|u| + C$
4. $\int e^u du = e^u + C$
5. $\int a^u du = \dfrac{a^u}{\ln a} + C \quad (a > 0,\ a \neq -1)$
6. $\int \sin u du = -\cos u + C$
7. $\int \cos u du = \sin u + C$
8. $\int \sec^2 u du = \tan u + C$
9. $\int \csc^2 u du = -\cot u + C$
10. $\int \sec u \tan u du = \sec u + C$
11. $\int \csc u \cot u du = -\csc u + C$
12. $\int \dfrac{1}{\sqrt{1-u^2}} du = \sin^{-1} u + C$
13. $\int \dfrac{1}{1+u^2} du = \tan^{-1} u + C$
14. ~~$\int \sinh u du = \cosh u + C$~~
15. ~~$\int \cosh u du = \sinh u + C$~~

- $\dfrac{d}{dx}\tan x = \sec^2 x$
- $\dfrac{d}{dx}\cot x = -\csc^2 x$
- $\dfrac{d}{dx}\sec x = \sec x \tan x$
- $\dfrac{d}{dx}\csc x = -\csc x \cot x$

$\int x^r\,dx = \frac{x^{r+1}}{r+1} \quad (r \neq -1) + C$

Example 5.22. Evaluate $\int (10x^4 - 2\sec^2 x)\,dx.$

rule 2

$= \frac{10x^5}{5} - 2\tan x + C$

Example 5.23. Evaluate $\int_{\pi/4}^{\pi/2} \frac{\cos\theta}{\sin^2\theta}\,d\theta.$

Quotient rules not applicable

$= \int \frac{\cos\theta}{\sin\theta} \cdot \frac{1}{\sin\theta}\,d\theta$

$= \int \cot\theta \cdot \csc\theta\,d\theta$

$= -\csc\theta + C$

- $\frac{d}{dx}\tan x = \sec^2 x$
- $\frac{d}{dx}\cot x = -\csc^2 x$
- $\frac{d}{dx}\sec x = \sec x \tan x$
- $\frac{d}{dx}\csc x = -\csc x \cot x$

Example 5.24. Evaluate $\int_0^3 (x^3 - 6x)\,dx.$

$= \left[\frac{x^4}{4} - \frac{6x^2}{2}\right]_0^3$

$= \left[\frac{x^4}{4} - 3x^2\right]_0^3$

$= \frac{3^4}{4} - 3\times 3^2$

$= \frac{81}{4} - 27 = -6.75$

Example 5.25. Evaluate $\displaystyle\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx.$

$$= \left[\frac{2x^4}{4} - \frac{6x^2}{2} + \frac{3}{2x}\ln|x^2+1|\right]_0^2$$

Example 5.26. Evaluate $\displaystyle\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2} dt.$

$$= \int_1^9 2 + \sqrt{t} - \frac{1}{t^2}\, dt$$

$$= \left[2t + \frac{2}{3}\cdot t^{\frac{3}{2}} + t^{-1}\right]_1^9$$

$$= \left[2\times 9 + \frac{2}{3}\times 9^{\frac{3}{2}} + 9\right] - \left[2 + \frac{2}{3} + 1\right]$$

$$= 41.333$$

Exercises 5.4 1, 3, 9, 13, 17, 23, 25, 27, 29, 33, 37, 41, 45, 59, 61.

5.5 The Substitution Rule ← chain rule

- **Substitution Rule for Indefinite Integrals** Apply to product of 2 functions

Let g be a differentiable function and suppose that F is an antiderivative of f, then

$$\int f\left(g(x)\right) g'(x)dx = F\left(g(x)\right) + C$$

Example 5.27. Evaluate $\int x^3\sqrt{x^4+11}dx$. ($\frac{d}{dx}$)

Let $u = x^4+11$; $\frac{du}{dx} = 4x^3 \rightarrow du = 4x^3dx$

$dx = \frac{du}{4x^3}$

$$\int x^3\sqrt{u}\,dx$$

$$= \int x^3\sqrt{u}\cdot\frac{1}{4x^3}du$$

$$= \frac{1}{4}\int\sqrt{u}\,du$$

$$= \frac{1}{4}\cdot\frac{2}{3}\cdot u^{\frac{3}{2}} + C$$

$$= \frac{1}{6}\cdot u^{\frac{3}{2}} + C$$

$$= \frac{(x^4+11)^{\frac{3}{2}}}{6} + C$$

Example 5.28. Evaluate $\int \sqrt{x^2+x}(2x+1)dx.$

$\frac{d}{dx}(x^2+x) = 2x+1$ ~ u.

Let $u = x^2+x$; $\frac{du}{dx} = (x^2+x)' = 2x+1$

$dx = \frac{du}{2x+1}$

$= \int \sqrt{u}\,(2x+1)\cdot \frac{1}{2x+1}\,du$

$= \int \sqrt{u}\,du$

$= \frac{2}{3}\cdot u^{\frac{3}{2}} + C$

$= \frac{2}{3}\cdot (x^2+x)^{\frac{3}{2}} + C$

Example 5.29. Evaluate $\int \frac{x^5}{\sqrt{x^2+1}}dx.$

$x^2 = u-1$

$x^4 = (u-1)^2$

Let $u = x^2+1$

$\frac{du}{dx} = 2x \rightarrow dx = \frac{du}{2x}$

$= \int x^5\cdot \frac{1}{\sqrt{x^2+1}}\,dx$

$= \int x^5\cdot \frac{1}{\sqrt{u}}\cdot \frac{1}{2x}\,du$

$= \frac{1}{2}\int x^4\cdot \frac{1}{\sqrt{u}}\cdot du$ ∵ 적분변수가 u 이므로 x 처리해줘야 함

$= \frac{1}{2}\int (u-1)^2\cdot \frac{1}{\sqrt{u}}\,du$

$= \frac{1}{2}\int (u^2-2u+1)\cdot u^{-\frac{1}{2}}\,du$

$= \frac{1}{2}\int u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + u^{-\frac{1}{2}}\,du$

$= \frac{1}{2}\left[\frac{2}{5}\cdot u^{\frac{5}{2}} - \frac{4}{3}u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right] + C$

$= $

$(u^{\frac{1}{2}})' = \frac{1}{2}u^{-\frac{1}{2}}$

Example 5.30. Find $\int \sin 3x dx$.

use comp-ize rule

let $u = 3x$; $\frac{du}{dx} = 3$ $dx = \frac{du}{3}$

$\int \sin 3x\, dx = \int \sin u \cdot \frac{1}{3} du$

$= \frac{1}{3} \cdot (-\cos u) + C$

$= -\frac{\cos 3x}{3} + C$

$= -\frac{\cos 3x}{3} + C$

try

Example 5.31. Evaluate $\int x \sin x^2 dx$.

let $u = x^2 \rightarrow \frac{du}{dx} = 2x \rightarrow dx = \frac{du}{2x}$

$\hookrightarrow x =$

$\int x \sin x^2 dx = \int x \sin u \cdot \frac{1}{2x} du$

$= \int \sin u\, du$

$= -\cos u + C$

$= -\cos x^2$

try

Example 5.32. Evaluate $\int \sin^3 2x \cos 2x dx$.

let $u = \sin 2x$; $\frac{du}{dx} = 2\cos 2x$, $dx = \frac{du}{2\cos 2x}$

$\int \sin^3 2x \cos 2x\, dx = \int u^3 \cdot \cos 2x \cdot \frac{1}{2\cos 2x} du$

$= \frac{1}{2}\int u^3 \cdot du$

$= \frac{u^4}{8} + C$

$= \frac{(\sin 2x)^4}{8} + C$

Example 5.33. Evaluate $\int \tan x dx$. 沒有 tanx 的 integration rule!

$= \int \frac{\sin x}{\cos x} dx$

Let $u = \cos x$; $\frac{du}{dx} = -\sin x$ $\rightarrow$ $dx = -\frac{1}{\sin x} du$

$= \int \frac{\sin x}{u} \cdot \left(-\frac{1}{\sin x}\right) du$

$= -\int \frac{1}{u} du$

$= -\ln|u| + C$

$= -\ln|\cos x| + C$ or $\ln|\cos x|^{-1} + C$

$= \ln|\sec x| + C$

Example 5.34. Evaluate $\int \sec x dx$.

- $\frac{d}{dx}\tan x = \sec^2 x$
- $\frac{d}{dx}\cot x = -\csc^2 x$
- $\frac{d}{dx}\sec x = \sec x \tan x$
- $\frac{d}{dx}\csc x = -\csc x \cot x$

$= \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$

$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$

Let $u = \sec x + \tan x$; $\frac{du}{dx} = \sec x \tan x + \sec^2 x$ $\rightarrow$ $dx = \frac{du}{\sec x \tan x + \sec^2 x}$

$= \int \frac{\sec^2 x + \sec x \tan x}{u} \cdot \frac{du}{\sec^2 x + \sec x \tan x}$

$= \int \frac{1}{u} du$

$= \ln|u| + C$

$= \ln|\sec x + \tan x| + C$

Example Evaluate $\int \sec x dx$

$= \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$

$= \int \frac{1}{\sec x + \tan x} d(\sec x + \tan x)$

$= \ln|\sec x + \tan x| + C$

- **Substitution Rule for Definite Integrals**

here upper and lower limit result is a constant

Let g have continuous derivative on $[a, b]$, and let f be continuous on the range of g, then

$$\int_a^b f\left(g(x)\right) g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

where $u = g(x)$.

Example 5.35. Evaluate $\int_0^4 \sqrt{2x+1}dx$.

method 1

Let $u = 2x+1$; $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

$\int_0^4 \sqrt{2x+1}\,dx = \int_{u_1}^{u_2} u^{\frac{1}{2}} \frac{1}{2}\,du$

$u_1 = 2\times 4+1 = 9$

$u_2 = 2\times 0+1 = 1$

$= \int_1^9 u^{\frac{1}{2}} \left(\frac{1}{2}\right) du$

$= \frac{1}{3}\left[u^{\frac{3}{2}}\right]_1^9$ 不用換回 x

$= \frac{1}{3}\left[9^{\frac{3}{2}} - 1\right] = \frac{26}{3}$

method 2

Let $u = 2x+1$, $\frac{du}{dx} = 2$

$\int u^{\frac{1}{2}} \cdot \frac{1}{2}\,du = \frac{1}{2}\cdot\frac{2}{3}u^{\frac{3}{2}} + C$

$= \frac{1}{3}\cdot u^{\frac{3}{2}} + C$

$= \frac{(2x+1)^{\frac{3}{2}}}{3} + C$

$\int_0^4 \sqrt{2x+1}\,dx = \frac{1}{3}\left[(2x+1)^{\frac{3}{2}}\right]_0^4$

$= \frac{1}{3}\left[9^{\frac{3}{2}} - 1\right]$

$= \frac{26}{3}$

Example 5.36. Evaluate $\int_0^1 \frac{x+1}{(x^2+2x+6)^2}dx$.

Let $u = x^2+2x+6$; $\frac{du}{dx} = 2x+2$

$dx = \frac{du}{2x+2}$

when $x = 1$, $u_1 = 1+2+6 = 9$

$x = 0$, $u_2 = 6$

$\int_0^1 \frac{x+1}{(x^2+2x+6)^2}\,dx = \int_6^9 \frac{x+1}{u^2}\cdot\frac{du}{2(x+1)}$

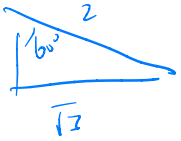
$= \frac{1}{2}\int_6^9 \frac{1}{u^2}\,du$

$= \frac{1}{2}\left[-u^{-1}\right]_6^9 = \frac{1}{-18} - \frac{1}{12} =$

Try

Example 5.37. Evaluate $\int_{\frac{\pi^2}{9}}^{\frac{\pi^2}{4}} \frac{\cos\sqrt{x}}{\sqrt{x}}dx.$

Let $u = \sqrt{x}$; $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$, $dx = 2\sqrt{x}\,du$



when $x = \frac{\pi^2}{9}$, $u = \frac{\pi}{3}$

$x = \frac{\pi^2}{4}$, $u = \frac{\pi}{2}$

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\cos u}{\sqrt{x}} \cdot 2\sqrt{x}\,du = 2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos u\,du = 2\left[\sin u\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= 2\left(\sin\tfrac{\pi}{2} - \sin\tfrac{\pi}{3}\right)$$

$$= 2 - \sqrt{3}$$

Try

Example 5.38. Evaluate $\int_1^e \frac{\ln x}{x}dx.$

Let $u = \ln x$, $\frac{du}{dx} = \frac{1}{x}$, $dx = x\,du$

when $x = e$, $u = \ln e = 1$

$x = 1$, $u = 0$

$$\int_0^1 \frac{u}{x} \cdot x\,du = \int_0^1 u\,du = \frac{1}{2}\left[u^2\right]_0^1$$

$$= \frac{1}{2}(1 - 0) = \frac{1}{2}$$

$$= \int_1^e \ln x\, d\ln x$$

$$= \frac{1}{2}\left[(\ln x)^2\right]_1^e$$

Example Evaluate $\int_1^e \frac{\ln x}{x}dx.$

$= \int_1^e \ln x\, d\ln x$

$= \frac{(\ln x)^2}{2}\Big|_1^e$

$= \frac{1}{2}(1^2 - 0)$

$= \frac{1}{2}$

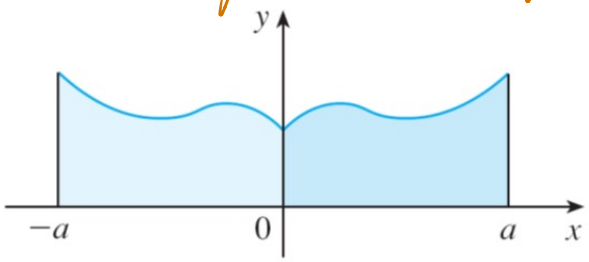
- **Symmetry Theorem**

If f is an even function, then

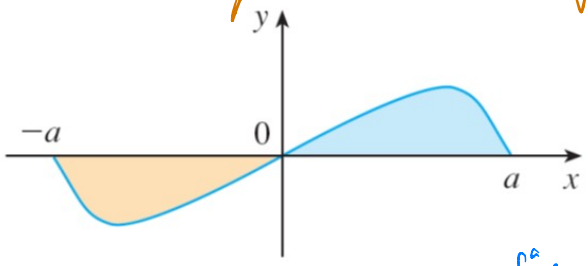
$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$$

If f is an odd function, then

$$\int_{-a}^{a} f(x)dx = 0$$



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

Example 5.39. Evaluate $\int_{-\pi}^{\pi} \cos\left(\frac{x}{4}\right) dx$.

Example 5.40. Evaluate $\int_{-5}^{5} \frac{x^5}{x^2+4} dx.$

Let $f(x) = \frac{x^5}{x^2+4}$

$f(-x) = -\frac{x^5}{x^2+4} = -f(x)$

$f(x)$ is an odd function

$\int_{-5}^{5} \cdots dx = 0$

Example 5.41. Evaluate $\int_{-2}^{2} \left(x \sin^4 x + x^3 - x^4\right) dx.$ (odd, odd, even)

$= -2\int_{0}^{2} x^4 \, dx$

$= -2 \cdot \frac{1}{5} \left[x^5\right]_0^2$

$= -\frac{2}{5}\left[2^5 - 0\right]$

$= -\frac{32}{5} = -\frac{64}{5}$

Example 5.42. Evaluate $\int_{-\pi}^{\pi} \sin^3 x \cos^5 x dx.$ (odd (even × odd = odd))

$= 0$

Try

Exercises 5.5 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 33, 35, 39, 41, 45, 47, 57, 59, 61, 65, 67, 69, 71, 73, 77, 81, 87, 91, 93.